

Abel–Ruffini via Pandrosion Quotient Orbits

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Abstract

The Abel–Ruffini theorem (1824) states that the general polynomial of degree ≥ 5 cannot be solved by radicals. We interpret this through the Pandrosion framework: the Galois group $\text{Gal}(P)$ acts on quotients $Q(\alpha_j, z)$ by permuting roots (Paper 74), and solvability by radicals requires this action to refine through abelian quotient steps. The obstruction is A_5 : a simple, non-abelian group whose derived series is [60] (stuck). We verify the solvability criterion, the discriminant test $\text{Gal} \subseteq A_n \iff \sqrt[n]{\Delta} \in \mathbb{Q}$, and confirm that $x^5 - x - 1$ ($\text{Gal} = S_5$) yields only implicit CRootOf solutions—no radical form exists.

1 The solvability criterion

Theorem 1.1 (Galois). *P is solvable by radicals if and only if $\text{Gal}(P)$ is a solvable group.*

n	$ S_n $	Solvable?	$ A_n $	Solvable?
2	2	✓	1	✓
3	6	✓	3	✓
4	24	✓	12	✓
5	120	×	60	×
6	720	×	360	×
7	5040	×	2520	×

The transition at $n = 5$: A_5 is *simple and non-abelian* (its derived series is [60], not reaching 1), blocking the composition series that exists for S_4 : $\{e\} \triangleleft V_4 \triangleleft A_4 \triangleleft S_4$ with abelian factors $\mathbb{Z}_2 \times \mathbb{Z}_2, \mathbb{Z}_3, \mathbb{Z}_2$.

2 Galois groups of specific polynomials

Polynomial	$ \text{Gal} $	Solvable	By radicals?
$x^2 - 2$	2 (S_2)	✓	Yes: $\pm\sqrt{2}$
$x^3 - 2$	6 (S_3)	✓	Yes: $\sqrt[3]{2} \cdot \zeta_3^k$
$x^4 - 2$	8 (D_4)	✓	Yes: $\sqrt[4]{2} \cdot i^k$
$x^5 - 2$	20 (M_{20})	✓	Yes: $\sqrt[5]{2} \cdot \zeta_5^k$
$x^5 - 5x + 12$	10 (D_5)	✓	Yes
$x^5 - x - 1$	120 (S_5)	×	No
$x^5 + 20x + 16$	60 (A_5)	×	No

Note: $x^5 - 2$ is solvable—its Galois group is M_{20} (order 20, solvable), not S_5 . The theorem says the *general* quintic is unsolvable, not every specific one.

3 The discriminant test (Paper 74)

Proposition 3.1. $\text{Gal}(P) \subseteq A_n \iff \Delta(P)$ is a perfect square in $\mathbb{Q}$.

Polynomial	Δ	Square?	Gal
$x^3 - 2$	-108	$\times$	S_3
$x^3 - 3x + 1$	$81 = 9^2$	$\checkmark$	$A_3 \cong \mathbb{Z}_3$
$x^5 - x - 1$	2869	$\times$	S_5
$x^5 + 20x + 16$	$1024000000 = 32000^2$	$\checkmark$	A_5

The case $x^5 + 20x + 16$ is instructive: Δ is a perfect square, so $\text{Gal} \subseteq A_5$, but A_5 is not solvable—the polynomial is *still* unsolvable by radicals.

4 Pandrosion interpretation

$\text{Gal}(P)$ acts on the Pandrosion quotients $Q(\alpha_j, z)$ by permutation (Paper 74). Solvability by radicals requires this action to decompose into a chain of “abelian steps”:

$$\{Q_1, \dots, Q_n\} \supset \text{orbit}_1 \supset \text{orbit}_2 \supset \dots \supset \{Q_i\}, \quad (1)$$

with each quotient group abelian. For S_5 : the quotient $S_5/A_5 \cong \mathbb{Z}_2$ is abelian, but $A_5/\{e\}$ is non-abelian—the chain terminates at A_5 and cannot be refined.

References

- [1] N. Abel, *Mémoire sur les équations algébriques*, 1824.
- [2] É. Galois, *Journal de Mathématiques*, 1846 (posthumous).
- [3] I. Besevic, *Galois theory of Pandrosion quotients*, Pandrosion Dynamics **74** (2026).